

Mobile Application Development with Flutter

Lecture 21: SizedBox & Card Widgets

Introduction

- In Flutter, controlling **spacing and layout structure** is very important for building clean and professional UI
- Two very useful widgets for this purpose are:
 - **SizedBox** → Used to create fixed spacing or fixed size widgets
 - **Card** → Used to create material design cards for displaying content
- Both widgets are very common in real world apps such as:
 - Product cards
 - User profile cards
 - Blog posts
 - Dashboard panels

SizedBox Widget

- SizedBox is a widget used to create **empty space** or a **box with fixed width and height**
- It is somewhat similar to a Container widget with fewer properties
- It helps in controlling layout spacing between widgets
- `SizedBox.expand()`
- `SizedBox.shrink()`
- `SizedBox.square()`
- `SizedBox.fromSize()`

SizedBox vs Container

SizedBox	Container
Lightweight	Heavy widget
Only size	Can style, padding, margin
Best for spacing	Best for styling

Card Widget

- A Card is a **Material Design widget** used to display content inside a **raised box with rounded corners and shadow**
- Used in:
 - Product lists
 - Profile cards
 - Blog previews
 - Dashboard panels

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Important Card Properties

Property	Description
elevation	Shadow depth
color	Background color
margin	Outer spacing
shape	Border shape
child	Content
shadowColor	Customizes shadow color

Best Practices

- Use SizedBox for spacing
- Always use Padding inside Card
 - Cards without padding look cramped
- Control elevation carefully
 - Too much elevation looks messy
 - Good range: 2 - 10